BABEŞ BOLYAI UNIVERSITY, CLUJ NAPOCA, ROMÂNIA
FACULTY OF MATHEMATICS AND COMPUTER SCIENCE

# AI-Enabled Cardiac Digital Twin from Wearable Sensors

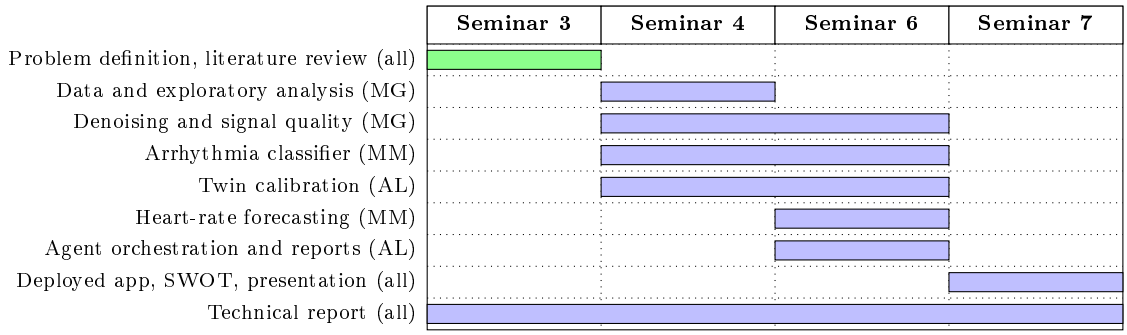
– Advanced Machine Learning report –

**Authors**
Marc Grec, [specialisation], [group], [email]
Mihai Moldovan, AI4CI, 270, mihai.moldovan2@stud.ubbcluj.ro
Adrien-Jeno Lorincz, [specialisation], [group], [email]

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## Team

| Member | Role | Milestone 1 tasks |
|---|---|---|
| Marc Grec, [email], [specialisation] | Signal Monitoring agent, data pipeline | Survey: wearable ECG, signal quality |
| Mihai Moldovan, `mihai.moldovan2@stud.ubbcluj.ro`, AI4CI | Arrhythmia Detection and Physiological Prediction agents | Survey: classification, forecasting |
| Adrien-Jeno Lorincz, [email], [specialisation] | Twin Calibration and Clinical Recommendation agents | Problem definition; survey: twins, agents |




Work plan (MG: Marc Grec, MM: Mihai Moldovan, AL: Adrien-Jeno Lorincz; green: done).

### Abstract

Wearables record a single-lead ECG continuously, but the signal is noisy and differs from person to person. We propose a simple cardiac digital twin: a model of the patient's heartbeat, fitted to their own normal beats and shared by five AI agents that clean the signal, detect arrhythmias, forecast heart rate, update the twin and write a report for a clinician. Beats that differ strongly from the twin are flagged as unusual for that patient. This report describes the problem (graphical abstract in Figure 2.1) and surveys related work. Experiments will use public PhysioNet ECG data.

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# 1 Introduction

Cardiovascular diseases are the leading cause of death worldwide, with an estimated 19.8 million deaths in 2022 [16]. Smartwatches can already screen people for heart-rhythm problems, but a single-lead wearable ECG is noisy and limited on its own [1]. Digital twins, models of one patient that are updated with that patient's data, are a promising way to personalise monitoring [3]. However, current cardiac twins are mostly built from hospital data such as medical imaging [6].

We build a simple twin from wearable data alone: a heartbeat model [10] fitted to the wearer's own beats, combined with machine-learning models that clean the signal, detect arrhythmias and forecast heart rate. Five agents share the twin and write a report for a clinician. The key idea is to measure how far each new beat is from the patient's own normal.

# 2 Scientific Problem

## 2.1 Problem definition

**Informal description.** Smartwatches and chest patches can record a single-lead ECG during everyday life, but these recordings are hard to use for three reasons:

- **Noise.** The person is moving, and wearable sensors have many sources of error, so the signal is often disturbed [1].
- **Volume.** A wearable records all day, every day, which is far more data than a clinician can read.
- **Every patient is different.** ECG shapes vary from person to person, so models trained on other people make more mistakes on a new patient [4, 8].

Our idea is a *cardiac digital twin*: a model of one patient's heart that is kept up to date with that patient's own data [3]. Existing cardiac twins are built from hospital imaging [6]; ours uses only wearable data. It is a simple mathematical model of a heartbeat [10], adjusted to the patient's own normal beats. When a new beat looks very different from what the twin expects, it is flagged as unusual for this patient. As in the course brief, five agents share the twin: Signal Monitoring, Arrhythmia Detection, Physiological Prediction, Twin Calibration and Clinical Recommendation. The result is a short report for a doctor; the system does not make diagnoses.

**Why an intelligent algorithm?** Fixed rules fail when the signal is noisy or the patient is atypical, while deep networks trained on large datasets classify heart rhythms from single-lead ECG as well as cardiologists [7]. Learning is also needed to clean the signal, forecast heart rate and fit the twin to each patient.

## 2.2 Formal problem statement

The input is a noisy ECG signal from a wearable, possibly together with heart rate and motion:

$$\tilde{x}(t) = x(t) + \eta(t), \tag{2.1}$$

where $x$ is the true heart signal and $\eta$ is the noise. From $\tilde{x}$ we cut out the individual heartbeats $b_1, b_2, \ldots$

The twin describes a heartbeat by its five waves (P, Q, R, S, T), each with a position, a height and a width, i.e. 15 numbers $\vartheta_p$ for patient $p$ [10]. Calibrating the twin means choosing $\vartheta_p$ so that the beats it generates match the patient's normal beats. For each new beat $b_k$ the twin predicts a beat $\hat{b}_k$, and the *deviation score*

$$d_k = \|b_k - \hat{b}_k\| \tag{2.2}$$

measures how unusual the beat is for this patient. A typical beat gives a small score, while a beat the twin cannot reproduce, for example an extra beat coming from the ventricles, gives a large one. Table 2.1 lists the input and output of each agent.

The agents work together as a pipeline (Figure 2.1). The Signal Monitoring agent cleans the signal and marks segments that are too noisy to use. The Arrhythmia Detection agent labels every beat, using the deviation score as an extra hint. The Physiological Prediction agent forecasts the heart rate. The Twin Calibration agent keeps the twin up to date, using only clean beats that were labelled normal. Finally, the Clinical Recommendation agent collects these results into a short report and highlights anything a doctor should look at.

Table 2.1: Input and output of each agent.

| Agent | Input | Output |
|---|---|---|
| Signal Monitoring | raw ECG | cleaned ECG and a signal-quality score |
| Arrhythmia Detection | heartbeats and their deviation scores | a label for every beat: normal (N), supraventricular (S), ventricular (V), fusion (F) or unknown (Q) [4] |
| Physiological Prediction | recent heart rate | heart-rate forecast |
| Twin Calibration | clean normal beats | updated twin $\vartheta_p$ and deviation scores $d_k$ |
| Clinical Recommendation | outputs of all agents | report for the clinician |

## 2.3 Graphical abstract

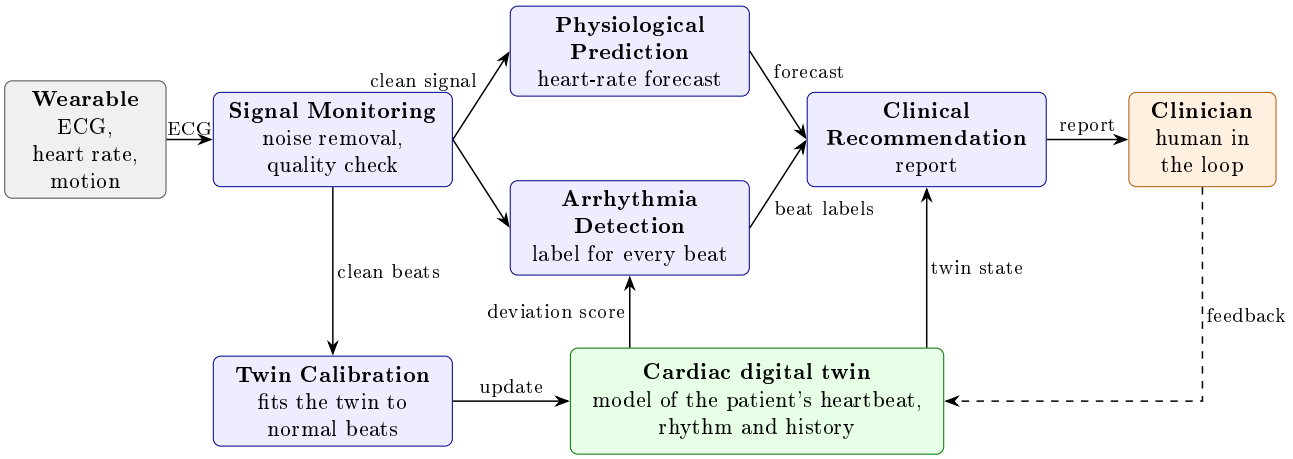



Figure 2.1: Graphical abstract: five agents share a model of the patient's heart (the twin); the clinician's feedback updates it.

# 3 State of the art/Related work

We searched Crossref, Semantic Scholar, OpenAlex, PubMed and the web for work in the areas below.

## 3.1 Cardiac digital twins

Corral-Acero et al. [3] define a healthcare digital twin as a model that combines a person's data over time, with wearables as one of its data sources. They argue that combining physiological models with statistical models learned from data is the way towards precision cardiology.

Most implemented cardiac twins use hospital data. Gillette et al. [6] build the heart's anatomy from MR images in a few hours per patient and fit its electrical behaviour to a 12-lead ECG, which they demonstrated for a single subject. Simonian et al. [14] build a twin of the mitral valve that integrates

the patient's follow-up data to predict the valve's state after a repair procedure. Thangaraj et al. [15] describe hybrid twins that combine physiological models with lightweight machine learning. They see smartwatches as a way to close the loop with the patient, but note that this only works if people keep wearing them. Closest to our work, Roop et al. [12] model the heart's electrical conduction system and personalise it with wearable signals, so that it produces ECGs that match the individual.

**Difference.** Our twin needs only a wearable, and it is used for continuous monitoring rather than simulation.

## 3.2 Wearable ECG and signal quality

Bayoumy et al. [1] review wearables in cardiovascular care. They find that a single-lead ECG is enough only for the simpler arrhythmias, and list accuracy, privacy, cost and separating useful information from noise as the main barriers to adoption. Bhavsar et al. [2] treat wearable measurements as digital biomarkers and propose a platform that connects patients and doctors and adds insights from continuous monitoring to clinical reports. Del-Valle-Soto et al. [5] survey wearable health applications, describe how they enable real-time feedback and personalised interventions, and stress the importance of privacy and data security.

Noise is the main technical obstacle. Sameni et al. [13] place a mathematical model of the ECG, the same kind of model our twin is based on, inside a Kalman filter that follows the signal over time. This removed noise better than standard filters over a wide range of noise levels.

**Difference.** We judge noise removal by how much it helps arrhythmia detection, not on its own.

## 3.3 Arrhythmia classification

De Chazal et al. [4] split the MIT-BIH Arrhythmia Database into two groups of patients, one for training and one for testing, which became the reference way to evaluate beat classifiers. With this split, their classifier detected some beat types, especially supraventricular beats, poorly. Kiranyaz et al. [8] therefore train a small network for each patient, using common data plus the patient's own labelled beats. It is fast enough for real-time monitoring and, according to the authors, could run on a lightweight wearable device. Hannun et al. [7] trained a deep network on single-lead ambulatory ECG that recognises 12 rhythm classes as well as cardiologists. They see its role in triage, with results confirmed by a clinician.

**Difference.** Instead of training a model per patient, we give a general model the twin's deviation score, which needs no labelled data from the new patient.

## 3.4 Heart models and forecasting

McSharry et al. [10] proposed the heartbeat model behind our twin, originally to generate realistic synthetic ECG for testing signal-processing methods. The model is controlled by a few intuitive settings: the heart rate, how much it varies, and the shape of the P, Q, R, S and T waves.

For forecasting heart rate, the Temporal Fusion Transformer [9] predicts several steps ahead and shows which inputs matter most. It combines different kinds of input, such as fixed information and past measurements, and was more accurate than earlier methods on real-world datasets.

**Difference.** We use the heartbeat model as a personal reference for each patient, not to generate data.

## 3.5 AI agents in healthcare

PHIA [11] is a language-model agent that analyses wearable health data by writing code and looking up information. In an evaluation by experts, it answered 84% of factual numerical questions correctly, showing that agents can reason over personal wearable data when they use tools instead of guessing.

**Difference.** Our agents coordinate the models and write the report; the clinician makes the decisions.

## 3.6 Positioning

Table 3.1 compares our project with the closest works. None of them combines all five features.

Table 3.1: Comparison with related work (✓: stated in the cited work).

| Work | Single-lead / wearable | Personalised | Heart model | Deep learning | AI agents |
|---|---|---|---|---|---|
| Gillette et al. [6] | | ✓ | ✓ | | |
| Roop et al. [12] | ✓ | ✓ | ✓ | | |
| Sameni et al. [13] | ✓ | ✓ | ✓ | | |
| Kiranyaz et al. [8] | ✓ | ✓ | | ✓ | |
| Hannun et al. [7] | ✓ | | | ✓ | |
| Merrill et al. [11] | ✓ | | | | ✓ |
| **This project** | ✓ | ✓ | ✓ | ✓ | ✓ |

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